

# Ryan Camilleri

Observational cosmologist

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## Summary

I am a postdoctoral research fellow at The University of Queensland specialising in observational cosmology, with a focus on standard candle techniques using Type Ia supernovae and tests of beyond- $\Lambda$ CDM models. I have strong experience building end-to-end cosmology pipelines for data analysis, simulation, and parameter inference alongside statistical analysis and model testing. I am also proficient working with established cosmological software frameworks such as COSMOSIS, SNANA, and COBAYA.

## Education

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|----------------------|---------------------------------------------------------------------------------------|----------------------------------------------|
| <b>PhD</b>           | <b>The University of Queensland</b> , Supernova Cosmology with the Dark Energy Survey | Brisbane, Australia<br>June 2022 – Sept 2026 |
| <b>BSc (Honours)</b> | <b>The University of Queensland</b> , Physics                                         | Brisbane, Australia<br>Feb 2021 – June 2022  |
| <b>BSc</b>           | <b>Griffith University</b> , Physics & Mathematics                                    | Gold Coast, Australia<br>Feb 2017 – Dec 2020 |

## Awards

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| <b>Honourable Mention for Best Student Publication in Physics</b>                                                                                                | 2025 |
| Awarded at The University of Queensland                                                                                                                          |      |
| <b>Dark Energy Survey “Builder”</b>                                                                                                                              | 2025 |
| Recognition for contributions to the DES Collaboration                                                                                                           |      |
| <b>Best Presentation - non-invited Speaker</b>                                                                                                                   | 2024 |
| Awarded at the Frontiers and Gravitational physics conference in Portsmouth, England.                                                                            |      |
| <b>Head of School Award for Excellence in Science</b>                                                                                                            | 2020 |
| Awarded to the Griffith University student in their final year with the highest career cumulative GPA in the Bachelor of Science (GPA 7.0).                      |      |
| <b>Australian Institute of Physics Prize</b>                                                                                                                     | 2020 |
| Awarded to the most outstanding graduating student in a Physics major (GPA 7.0).                                                                                 |      |
| <b>Griffith Futures Scholarship</b>                                                                                                                              | 2020 |
| Awarded to students who have been impacted by various types of educational disadvantage.                                                                         |      |
| <b>Griffith Award for Academic Excellence</b>                                                                                                                    | 2017 |
| Awarded to students who achieve a cumulative GPA of at least 6.0, calculated over the academic year. I received this award each year of my undergraduate degree. |      |

## Conferences

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| <b>Next Generation of SN Surveys Mini-Workshop</b>                     | Oxford, United Kingdom<br>July 2026 |
| Presented my paper: Supernova Unite: Combining Pantheon+ and DES-SN5YR |                                     |
| <b>Supernova Cosmology: New Challenges and Opportunities</b>           | Seoul, South Korea<br>June 2026     |
| Presented my paper: Supernova Unite: Combining Pantheon+ and DES-SN5YR |                                     |

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| <b>The End of <math>\Lambda</math>?</b><br>Invited talk presenting both the DES supernova cosmology results and my current project, which is a combined DES & Pantheon+ analysis | Incheon, South Korea<br>Sept 2025 |
| <b>Cosmic Flows</b><br>Presented $H_0$ inverse distance ladder results                                                                                                           | Brisbane, Australia<br>Feb 2025   |
| <b>DES Spring Meeting</b><br>Presented beyond- $\Lambda$ CDM paper                                                                                                               | Barcelona, Spain<br>May 2024      |
| <b>Frontiers in Cosmology &amp; Gravitational Physics</b><br>Presented beyond- $\Lambda$ CDM paper                                                                               | Portsmouth, England<br>May 2024   |
| <b>OzGrav ECR Winter School</b><br>Attended the OzGrav ECR Winter School                                                                                                         | Kingscliff, Australia<br>Aug 2025 |

## Publications

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| <b>Supernova Unite: Constraints on cosmologies beyond-<math>\Lambda</math>CDM</b><br>In preparation<br>R. Camilleri et al.                                                              | 2026 |
| <b>Supernova Unite: Combining Pantheon+ and DES-SN5YR</b><br>Submitted to PASA<br>R. Camilleri et al.<br>(arXiv)                                                                        | 2026 |
| <b>DES Supernova Program: <math>H_0</math> from the Inverse Distance Ladder</b><br>R. Camilleri et al.<br>(MNRAS)                                                                       | 2025 |
| <b>DES Supernova Program: Investigating Beyond-<math>\Lambda</math>CDM</b><br>R. Camilleri et al.<br>(MNRAS)                                                                            | 2024 |
| <b>DES Supernova Program: Reanalysis of Cosmology Results</b><br>B. Popovic et al.<br>(MNRAS)                                                                                           | 2025 |
| <b>Dark Energy Survey: implications for cosmological expansion models from the final DES Baryon Acoustic Oscillation and Supernova data</b><br>DES Collaboration<br>(Physical Review D) | 2025 |
| <b>Cosmological non-linearities as an effective fluid</b><br>L. Giani, R. Camilleri et al.<br>(Physical Review Letters)                                                                 | 2025 |

## Activities

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| <b>UQ-QASMT Research Mentoring Program</b><br>I guided a group of Year 10 students through an independent cosmology project, introducing them to Python coding, and helping them develop the skills to produce clear plots and reports that effectively communicate their results. | July 2025 – 2026 |
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**Dark Energy Survey**

2022 – 2025

I organised and led tri-weekly international meetings, facilitating collaboration between Australia and UK teams.

**UQ Astro Work Experience Program**

2022 – present

The Astronomy group at The University of Queensland runs a weeklong work experience program (typically mid-year) and offers future STEM students who want careers in astronomy a first-hand look at the day-to-day life of an Astronomer. The week consists of an astronomy project, telescope demonstrations, “Meet the researcher” discussions and on the final day, the students get to present what they have learned to the group. Since 2022, I have volunteered to help organise and run this program.

## Memberships

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**Dark Energy Survey (DES):** Builder. The Dark Energy Survey is an international, collaborative effort to map hundreds of millions of galaxies, detect thousands of supernovae, and find patterns of cosmic structure that will reveal the nature of the mysterious dark energy that is accelerating the expansion of our Universe.

**ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav):** Member. The mission of OzGrav is to capitalise on the historic first detections of gravitational waves to understand the extreme physics of black holes and warped spacetime, and to inspire the next generation of Australian scientists and engineers through this new window on the Universe.

**Dark Energy Spectroscopic Instrument (DESI) Collaboration:** Member. DESI seeks to map the large-scale structure of the Universe over an enormous volume. Targeting about 30 million pre-selected galaxies across one-third of the night sky, scientists will use DESI’s redshifts data to construct 3D maps of the universe.

## References

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**Prof. Tamara Davis**

**A/Prof. Maria Vincenzi**

**Dr Paul Shah**